

The knob on one disk fits into the hole on the other.



Snap and pop

- **What?** Snaps
- **Who?** Heribert Bauer
- **Where and when?** Germany, 1885

Small interlocking disks were occasionally used as fasteners in ancient China, but the modern metal snap fastener was the work of the German inventor Heribert Bauer in the late 19th century. He called them *federknopf-verschluss*, which means “spring button closers.” They are often used for securing children’s clothes because they are very easy to open and close, but they also sometimes appear on adult clothing.

WOW!

The word “button” comes from an ancient Germanic word, *buttan*, meaning something that sticks out.

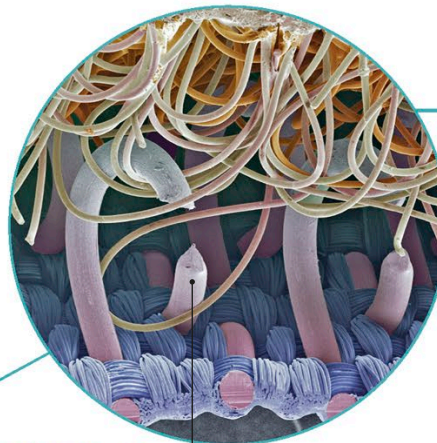
Modern
Velcro®

Pull it together

- **What?** Zipper
- **Who?** Gideon Sundback
- **Where and when?** Sweden, 1913

The first zipperlike fastener was the work of the American inventor Whitcomb Judson, in 1893. It consisted of a row of metal eyes and hooks (rather than the

metal teeth of modern zippers), pressed together by a slider. Called the “C-curity,” it was ingenious but it didn’t work very well. So, Judson hired the Swedish scientist Gideon Sundback to make improvements. In 1913, Sundback came up with the zipper we know today.



The tiny hooks on one surface cling to the tiny loops on the other.

Hooks and loops

- **What?** Velcro®
- **Who?** Georges de Mestrel
- **Where and when?** Switzerland, 1955

De Mestrel based his fastener on burrs—sticky seeds that are covered in tiny hooks to help them cling to rough surfaces, such as animal fur. He constructed it from two materials: one covered in tiny hooks and the other in tiny loops, which stick tight when pressed together. He got the name for his invention, Velcro®, from the French words *velours croché*, meaning “hooked velvet.”